

Jialong Fan

Third-Year Undergraduate, Physics + Computer Science, The Hong Kong University of Science and Technology |
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PhD Research Interests

Prospective PhD applicant interested in quantum information, quantum computing, quantum error correction, and fault-tolerant quantum computation. Current preparation centers on quantum LDPC codes, toric codes, BP-OSD decoders and variants, high-dimensional expanders, and quantum locally testable codes.

Education

The Hong Kong University of Science and Technology, Undergraduate Student in Physics + Computer Science 2023 - Present

- Third-year undergraduate with cross-disciplinary training in physics and computer science.
- Academic interests lie at the interface of quantum theory, computational methods, algorithms, and mathematical structures in quantum information.

Research Preparation and Academic Experience

Study in Quantum Error Correction and FTQC, Shilin Huang Group Junior Year - Present

- Studying fault-tolerant quantum computation and quantum error correction topics with exposure to qLDPC codes, toric codes, BP-OSD decoders, and related variants.
- Developing a theoretical foundation for code constructions, decoding algorithms, and their roles in scalable fault-tolerant architectures.

UROP Learning Experience in NV-Center Quantum Computing, Yang Sen Group, HKUST Sophomore Fall

- Gained introductory exposure to NV-center-based quantum computing through an HKUST UROP project, building context for physical platforms and quantum information applications.

Independent Study and Research Direction

- Independently studying high-dimensional expanders and their connections to quantum locally testable codes, with attention to locality, testability, distance, rate, and decodability.
- Particularly drawn to theoretical, mathematical, and abstract problems in quantum error correction and quantum information.
- Comfortable learning across physics, mathematics, and computer science, and motivated to connect formal theory with simulation and algorithmic understanding.

Technical Skills

Programming	Python; basic C++; R.
Quantum Simulation	Experience using the Stim Python library for quantum-circuit and quantum-error-correction simulations.
Relevant Topics	FTQC; qLDPC codes; toric codes; BP-OSD decoders and variants; high-dimensional expanders; qLTC.

Language

TOEFL iBT: 113

Note: Publications, honors, teaching, conference presentations, and references are not listed at this stage and can be added for program-specific versions.